AN OPEN IMMERSION OF ALGEBRAIC STACKS IS ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The trivial implications among properties of morphisms of algebraic stacks, in particular that a representable morphism is DM. Invoked inside the counted proof (source lines 8121-8122). Gives that the open immersion is DM.
- The statement for schemes that an open immersion is etale. Invoked inside the counted proof (source lines 8125-8126) as a cross-chapter pointer. Provides etaleness of the induced morphism of presenting schemes.

Source reference. Original lemma lemma-trivial-implications: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4773–4773. Navigation pointer only: the counted text refers to the Stacks item lemma-trivial-implications, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference morphisms-lemma-open-immersion-etale: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8126–8126. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-open-immersion-etale, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *An open immersion is étale.*

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Proof. Let $j : \mathcal{U} \to \mathcal{X}$ be an open immersion of algebraic stacks. Since j is representable, it is DM by Lemma lemma-trivial-implications. On the other hand, if $X \to \mathcal{X}$ is a smooth and surjective morphism where X is a scheme, then $U = \mathcal{U} \times_{\mathcal{X}} X$ is an open subscheme of X. Hence $U \to X$ is étale (Morphisms, Lemma morphisms-lemma-open-immersion-etale) and we conclude that j is étale from the definition. □